Chittagong University of Engineering & Technology

Department of Electronics and Telecommunication Engineering

EXPERIMENT NO: 05

Name of the Experiment

Implementation of RIP for Dynamic Routing in different network Topologies using CISCO routers

Course No : ETE 412

Course Title : Advanced Communications and Networking Sessional

Date of Experiment : 11/01/2026

Date of Submission : 25/01/2026

REMARKS

Objectives

• To implement dynamic routing using RIP protocol across multiple networks.
• To observe automatic route learning and end-to-end communication among PCs.

Software

• Cisco Packet Tracer

Introduction

Routing Information Protocol (RIP) is a dynamic routing protocol that uses distance-vector algorithm to determine the best path. Routers automatically exchange routing information and update routing tables periodically. RIP uses hop count as a metric and supports a maximum of 15 hops for packet forwarding.

Procedure

1. Network topology was created using routers, switches, and PCs.
2. IP addresses and subnet masks were assigned to all interfaces.
3. Default gateways were configured on all PCs.
4. Interfaces were activated using `no shutdown`.
5. RIP protocol was enabled using `router rip`.
6. Network statements were configured using `network` command.
7. Routing tables were verified using `show ip route`.
8. Connectivity was tested using ping.

Connection Diagram

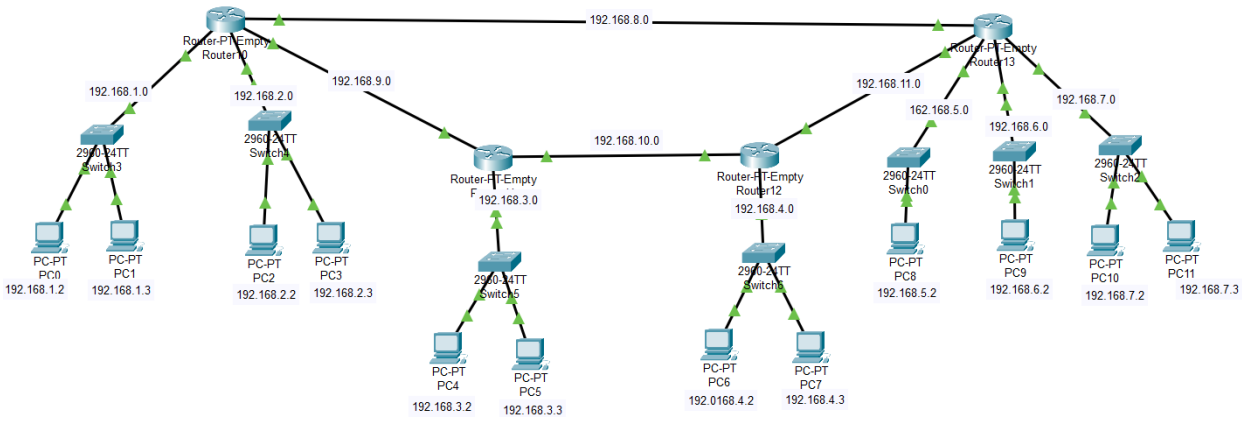
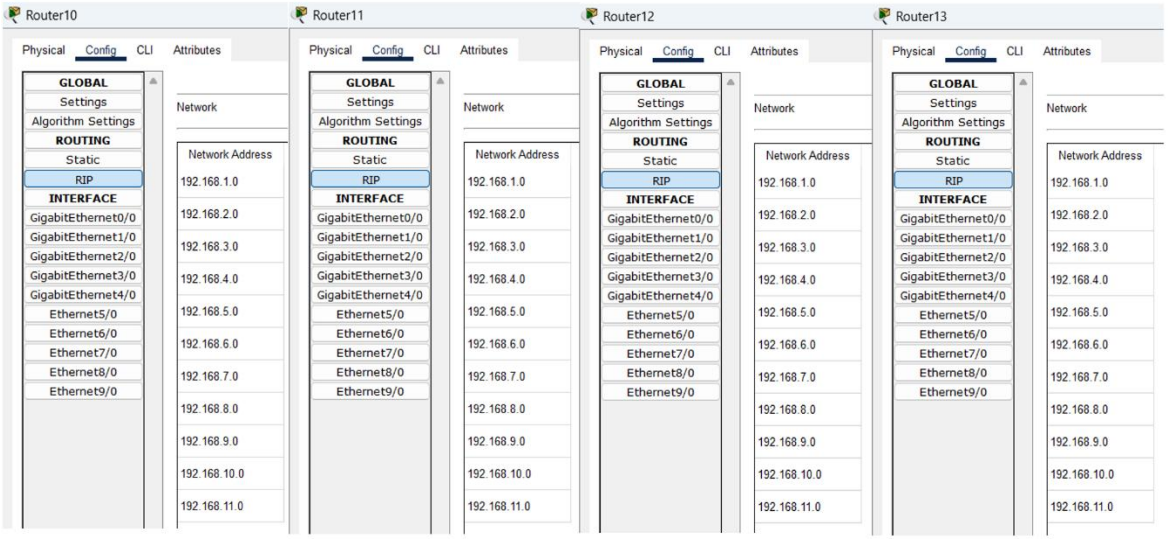


Figure 5.1 : Implementation of RIP dynamic routing in a network using multiple router-Topology 1, all the IP has the subnet mask **of 255.255.255.0**.

Subnet Mask: 255.255.255.0 (Applied to All Networks)			
Network Address	**Default Gateway**	**Network Address**	**Default Gateway**
192.168.1.0	192.168.1.1	192.168.7.0	192.168.7.1
192.168.2.0	192.168.2.1	192.168.8.0	192.168.8.1
192.168.3.0	192.168.3.1	192.168.9.0	192.168.9.1
192.168.4.0	192.168.4.1	192.168.10.0	192.168.10.1
192.168.5.0	192.168.5.1	192.168.11.0	192.168.11.1
192.168.6.0	192.168.6.1	----------	----------

(a) Default Gateway Allocation for Class C Networks (/24) network Topology 1.



(b) RIP routing address Allocation for Class C Networks (/24) network Topology 1.

Table 5.1 : Network Addressing Tables for RIP Topology 1.

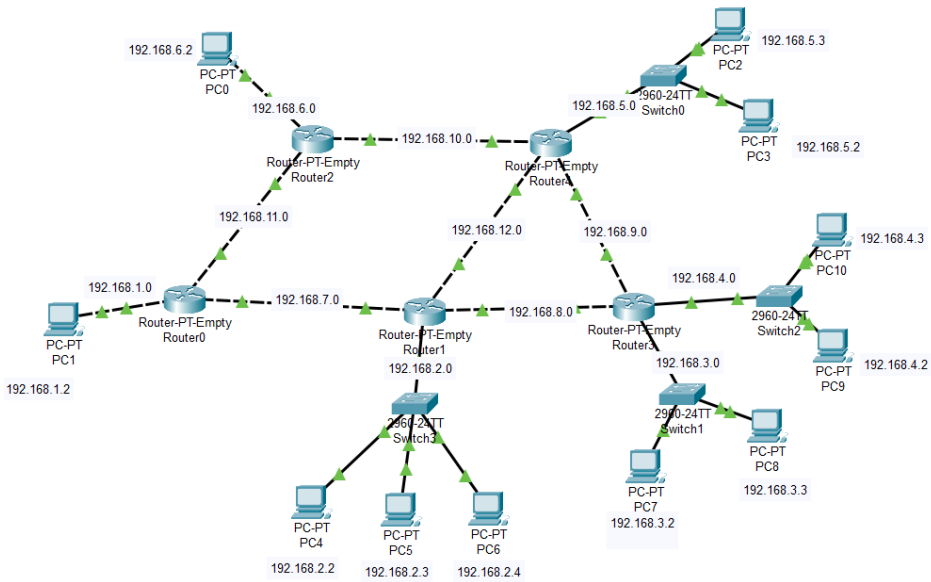
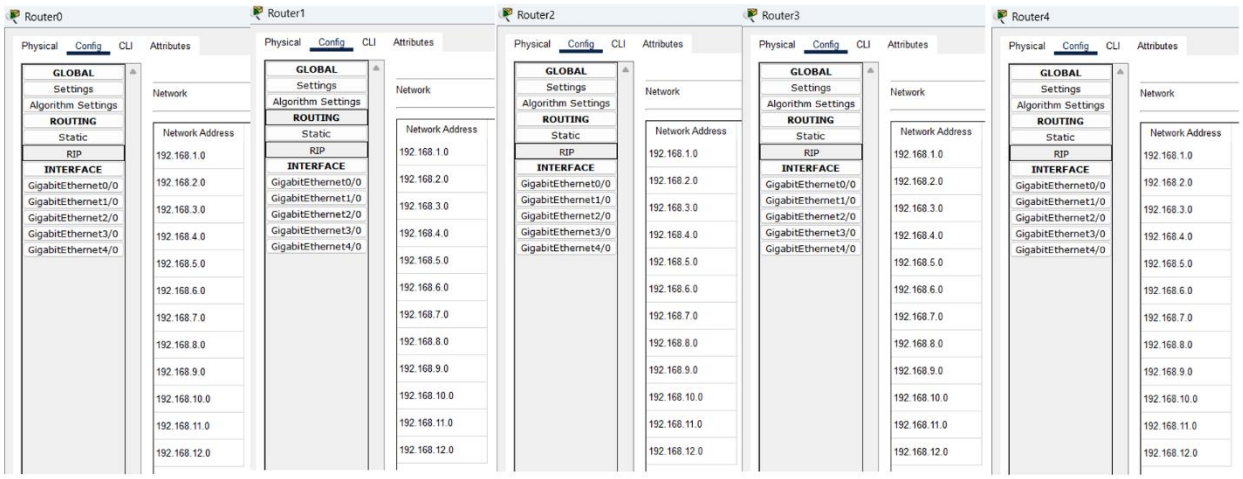


Figure 5.2 : Implementation of RIP dynamic routing in a network using multiple routers Topology 2, all the IP has the subnet mask of **255.255.255.0**

Subnet Mask: 255.255.255.0 (Applied to All Networks)			
Network Address	**Default Gateway**	**Network Address**	**Default Gateway**
192.168.1.0	192.168.1.1	192.168.7.0	192.168.7.1
192.168.2.0	192.168.2.1	192.168.8.0	192.168.8.1
192.168.3.0	192.168.3.1	192.168.9.0	192.168.9.1
192.168.4.0	192.168.4.1	192.168.10.0	192.168.10.1
192.168.5.0	192.168.5.1	192.168.11.0	192.168.11.1
192.168.6.0	192.168.6.1	192.168.12.0	192.168.12.1

(a) RIP routing address Allocation for Class C Networks (/24) for network Topology 2.



(b) RIP routing address Allocation for Class C Networks (/24) Network Topology 2.

Table 5.2 : Network Addressing Table for RIP Topology 2.

Discussion

1. RIP protocol was configured to enable dynamic routing between networks.
2. Routing tables were updated automatically through periodic exchanges.
3. Hop count metric was used to determine the optimal routing path.
4. Manual route configuration was eliminated through dynamic learning.
5. Network convergence was achieved automatically after configuration.
6. End-to-end connectivity was verified using ICMP (ping) testing.